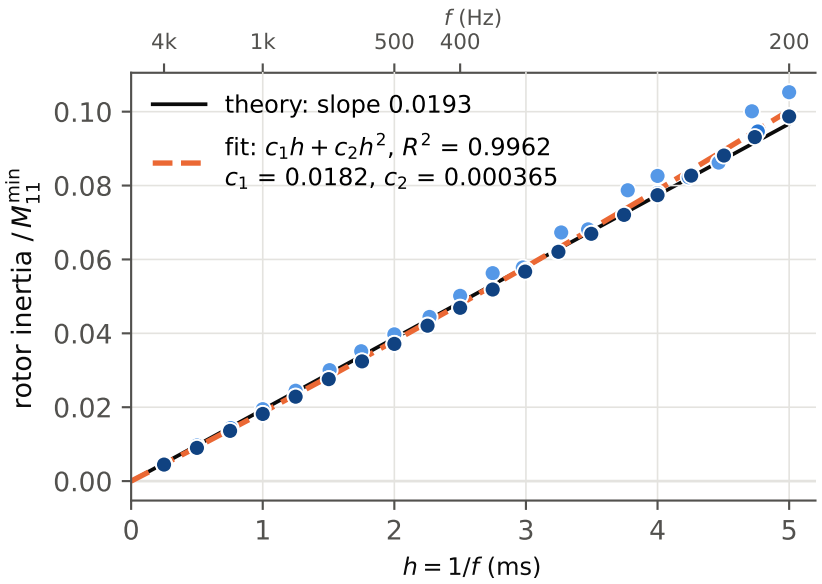
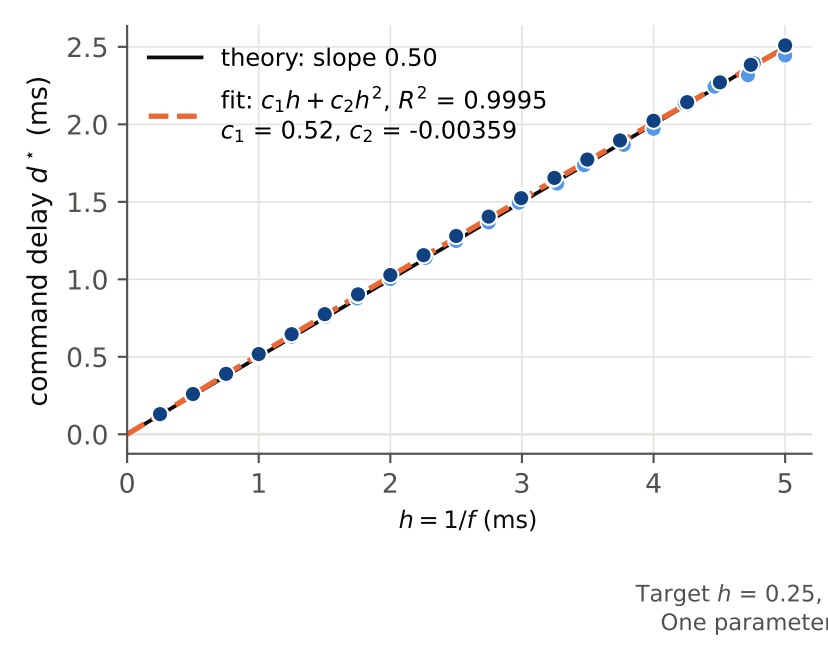
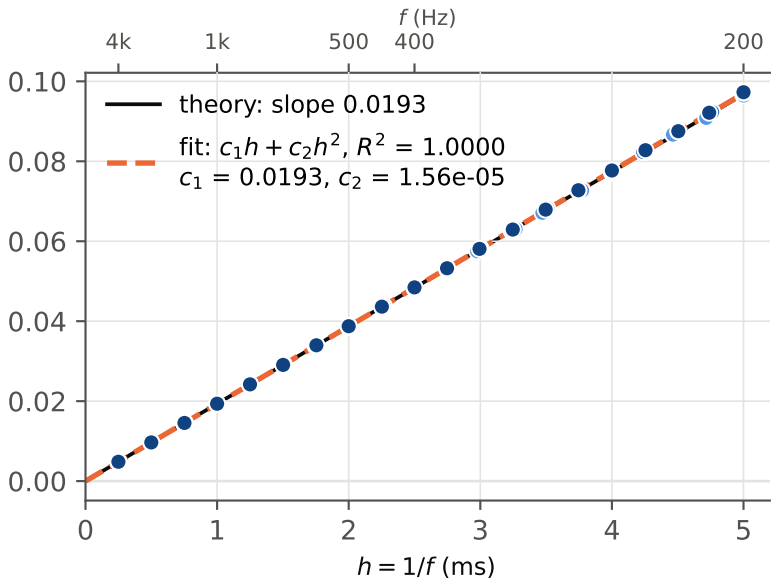
$f_{ctrl} \approx$ 50 Hz $f_{ctrl} \approx$ 100 Hz $f_{ctrl} \approx$ 200 Hz

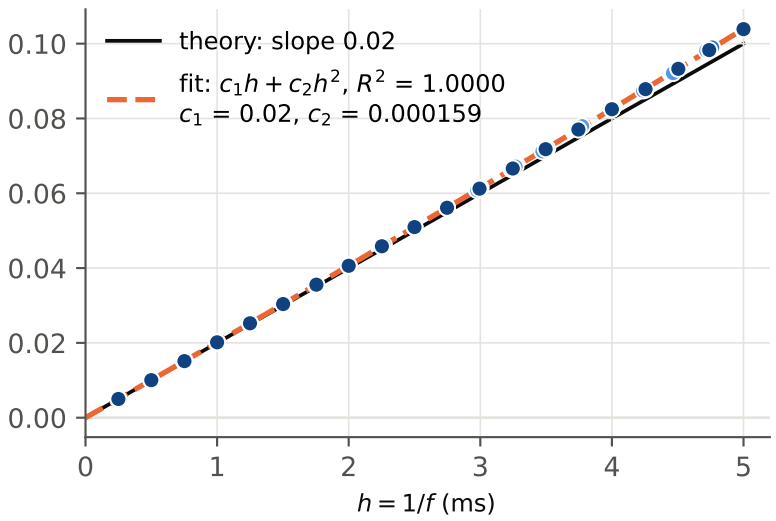
A: Euler + explicit PD
fitted: armature, K_D, d





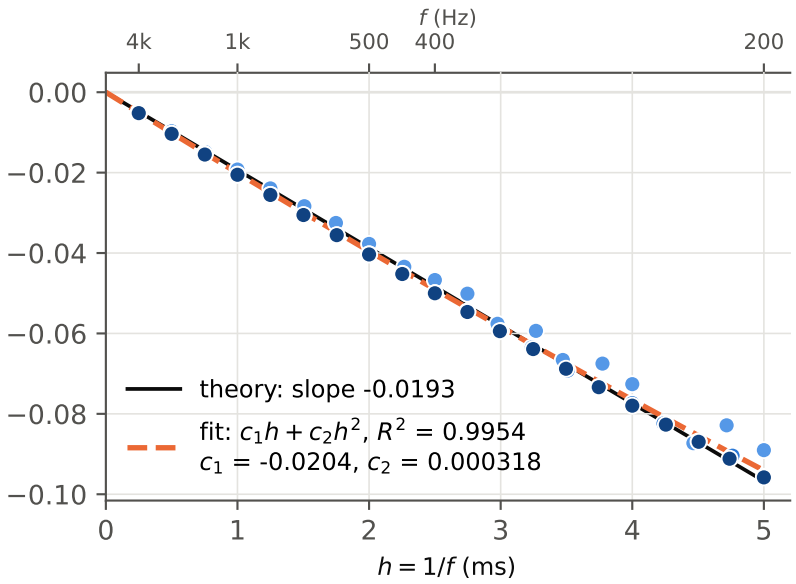
B: RK4 + explicit PD (torque held)
fitted: armature, K_D

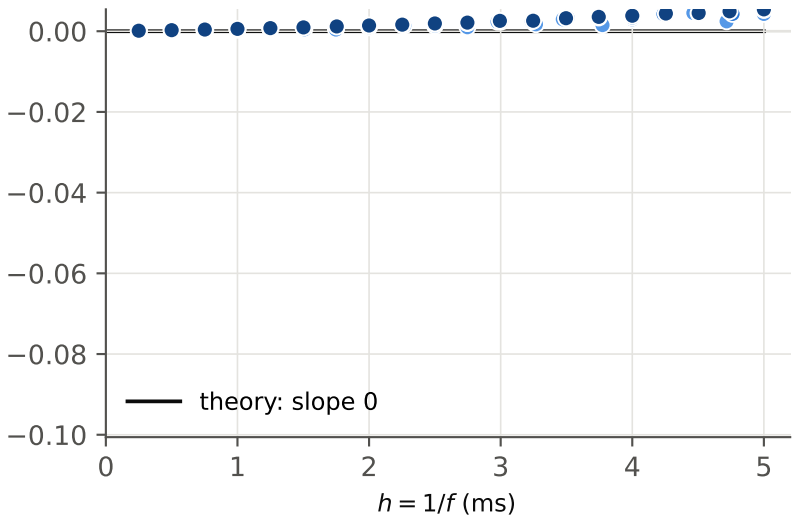


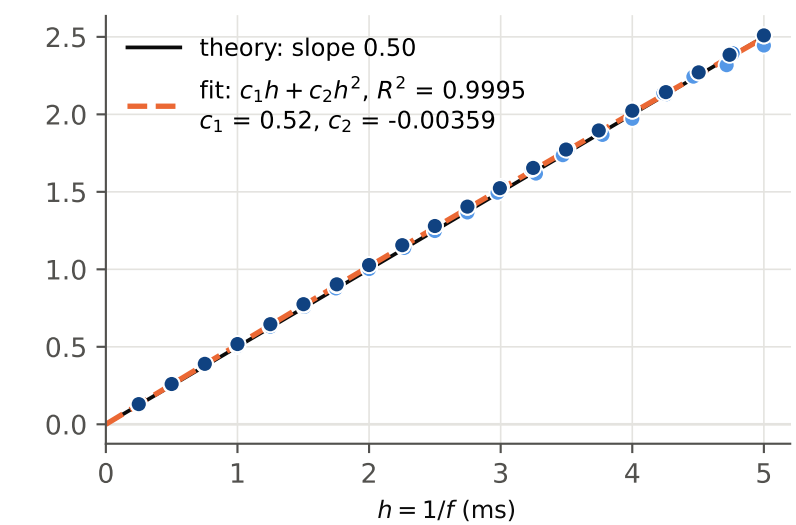


not fitted:
no command advance in this scheme

C: implicitfast + native PD
fitted: armature, K_D, d







Target h = 0.25, 0.5, ..., 5 ms (f = 200-4000 Hz), realized with integer f_{ctrl} (Hz) and integer decimation.
One parameter set is fitted to five drive frequencies at once, over 0.1 s, both joints in the residual.
K_P is held at its measured value.
First-order theory (solid); fit (dashed), $c_1h + c_2h^2$ through the origin, so c_1 is the measured first-order slope.